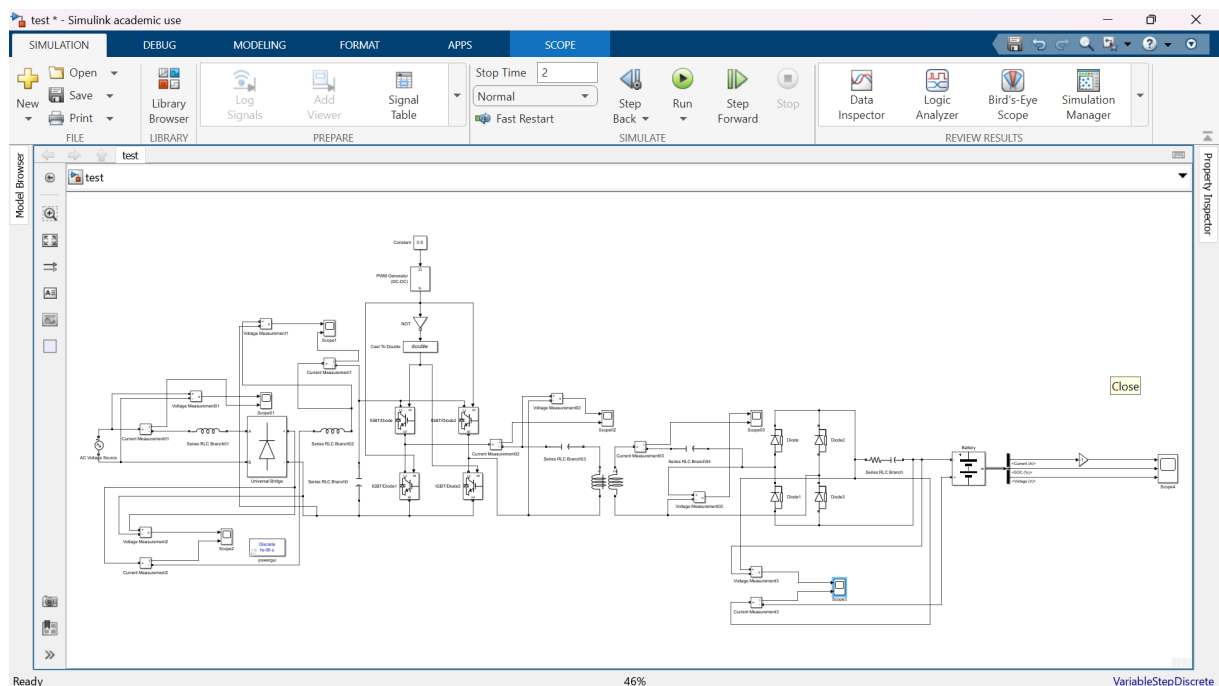


ROBUST INDUCTIVE WIRELESS ELECTRIC VEHICLE CHARGING

Resonant power transfer, engineering design, robustness analysis, and closed-loop power regulation

MATLAB Simulink · Simscape Electrical · Power Electronics · Magnetic Coupling · Control Engineering



Complete switching-level charging model implemented in Simulink and Simscape Electrical.

Prepared as an engineering project portfolio document

The report deliberately distinguishes measured switching-level simulation behavior from analytical design-study results so that every performance claim remains technically defensible.

1. Executive Summary

This project develops a resonant inductive wireless charging system for an electric vehicle using MATLAB Simulink and Simscape Electrical. The implemented model demonstrates the complete energy-transfer path: electrical source, high-frequency full-bridge inverter, primary resonant network, transmitter coil, magnetic air gap, receiver coil, secondary resonant network, rectifier, filtering stage, and battery model.

The engineering work goes beyond building the simulation. The transmitter and receiver resonant components were designed around a 30 kilohertz operating point, the nominal magnetic coupling was calculated from the mutual inductance, an equivalent coil-resistance loss budget was derived for a 90 percent efficiency target, and several compensation arrangements were compared. A separate robustness study evaluates air-gap variation, lateral displacement, frequency sensitivity, and closed-loop output-power regulation.

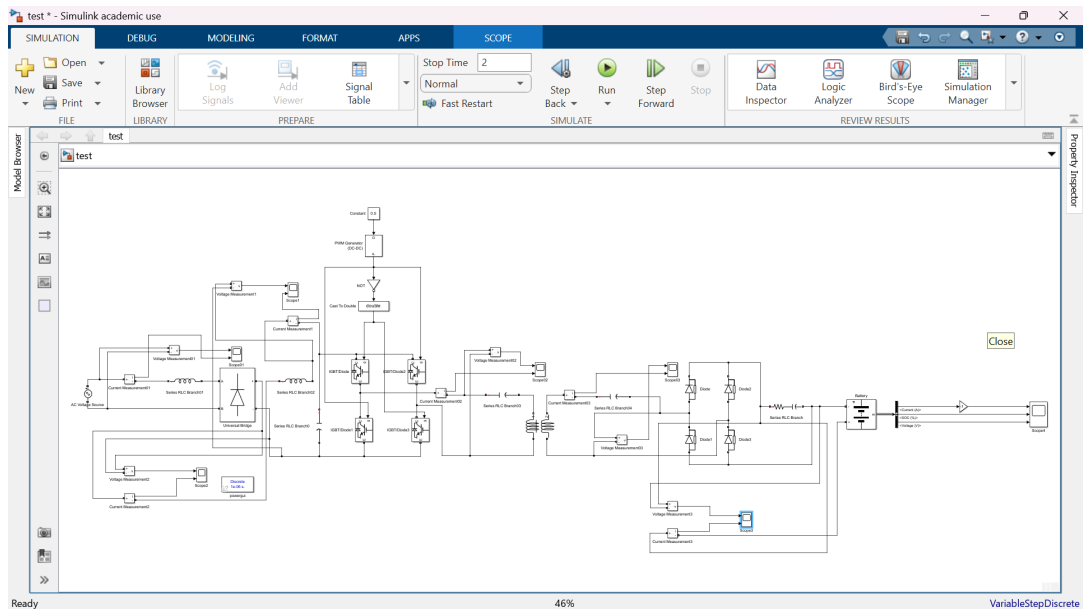
The central engineering question is: **how can a wireless charging system maintain controlled battery-side power when the magnetic coupling changes because the vehicle moves relative to the charging pad?** The answer developed here is a resonant series-series architecture combined with phase-shift control of the inverter.

The switching-level model successfully demonstrates wireless power transfer and battery-side charging behavior. The analytical design study then quantifies how the system responds to disturbances and how much control effort is required to maintain approximately 7.96 kilowatt output power.

2. Engineering Objectives

- Build a complete switching-level wireless charging model rather than a simplified block-level power-flow diagram.
- Design the transmitter and receiver compensation networks around a common 30 kilohertz resonant operating point.
- Establish a nominal operating condition near 390 volt, 20.4 ampere, and 7.96 kilowatt on the battery side.
- Quantify the magnetic coupling represented by the mutual inductance in the implemented model.
- Create an equivalent loss budget corresponding to a 90 percent design-efficiency target.
- Compare series-series, series-parallel, parallel-series, and parallel-parallel compensation arrangements.
- Study sensitivity to air gap, lateral displacement, and operating frequency.
- Develop and evaluate proportional-integral phase-shift regulation of battery-side output power.
- Document the assumptions and limitations clearly enough that the work can be extended to a more realistic charging-pad design.

3. System Architecture



System-level power-transfer architecture.

The implemented chain is intentionally modular. Each stage has a distinct engineering purpose and can be observed through voltage and current measurements. This makes it possible to determine whether an unexpected result originates in the inverter, resonant network, magnetic coupling, rectifier, filter, or battery interface.

Stage	Purpose
Electrical source	Provides the electrical input energy to the power converter.
Full-bridge inverter	Creates the high-frequency switched excitation used to drive the resonant transmitter.
Primary resonant network	Compensates the transmitter coil and establishes the intended resonant operating point.
Magnetic coupling	Transfers energy across the air gap without a direct electrical connection.
Secondary resonant network	Receives and conditions the magnetically transferred energy.
Diode bridge rectifier	Converts received alternating voltage and current into unidirectional power.
Output filter	Reduces rectified ripple before the battery interface.
Battery model	Represents the vehicle-side energy-storage load and exposes terminal voltage, current, and state of charge.

4. Why the Resonant Design Was Chosen

4.1 Operating frequency

A 30 kilohertz operating frequency was selected as the common design point. The objective was not simply to choose a frequency and then tune the model to it. The frequency establishes the required compensation capacitances for the selected coil inductances and determines the impedance behavior of the power-transfer link. Keeping the transmitter and receiver close to the same resonant frequency allows the magnetic link to transfer substantial power while reducing the reactive component of the coil impedance.

4.2 Transmitter and receiver inductances

The implemented model uses approximately 266 microhenry on the transmitter side and 257 microhenry on the receiver side. These values define the energy-storage capability of the resonant tanks and directly determine the compensation capacitances needed for the chosen frequency.

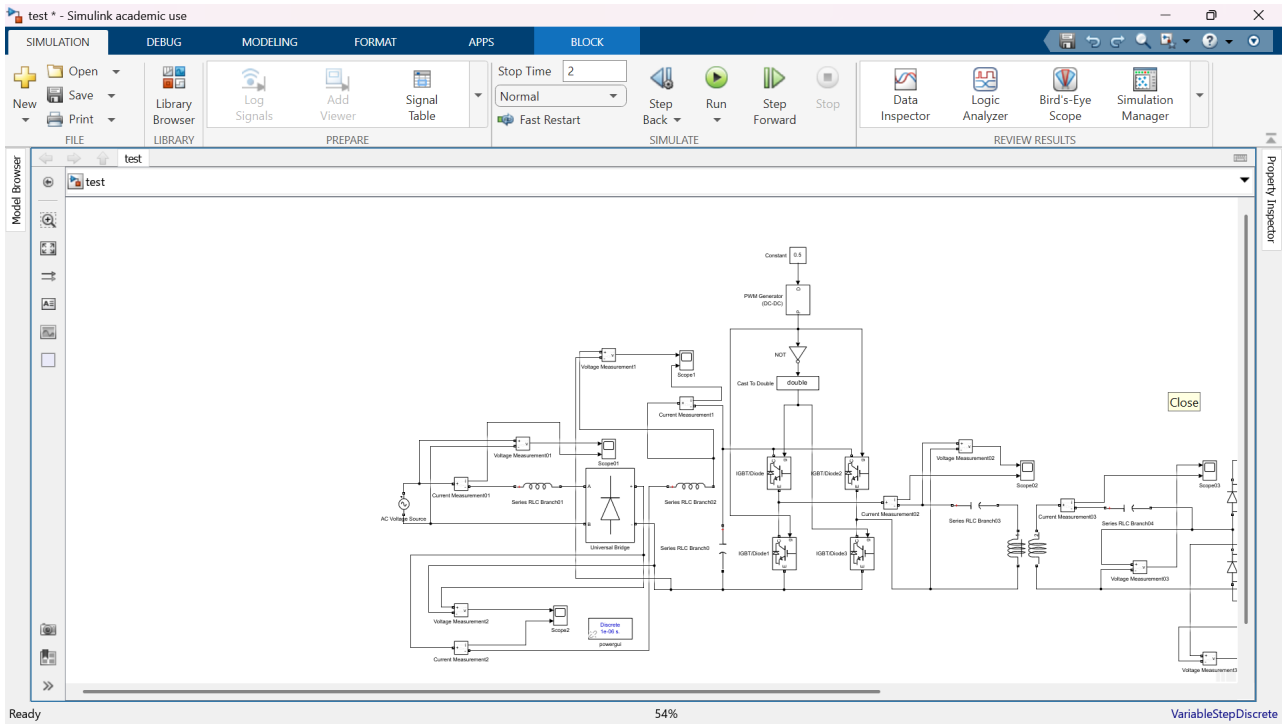
4.3 Compensation capacitances

Using the resonance relationship $f = 1 / (2\pi\sqrt{LC})$, the selected inductances lead to approximately 106 nanofarad on the transmitter and 110 nanofarad on the receiver. The resulting calculated resonant frequencies are approximately 29,973 hertz and 29,933 hertz, respectively. This small difference from 30 kilohertz is acceptable for the design study.

and also explains why the frequency sweep shows a slightly shifted best-efficiency point.

4.4 Series-series compensation

Series-series compensation was selected as the main topology because both coils are placed in direct series-resonant paths. The arrangement gives a straightforward energy-transfer path and moderate coil current at the nominal coupling. The later topology study confirms that this choice is not arbitrary: parallel-secondary arrangements produce much higher current in the studied low-coupling condition, while one alternative loses charging capability at a large air gap.



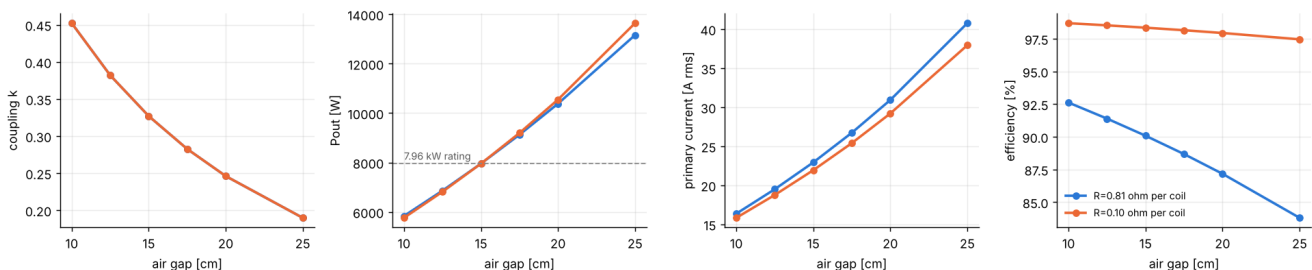
Inverter and resonant section of the implemented model.

5. Magnetic Coupling and Coil Separation

The magnetic link is represented using mutual inductance. The nominal mutual inductance is 85.46 microhenry. The corresponding coupling coefficient is calculated from $k = M / \sqrt{(L_1 L_2)}$, giving approximately 0.327.

This coupling coefficient is deliberately treated as a variable in the robustness study. Vehicle position changes the amount of magnetic flux shared by the two coils, which changes mutual inductance and therefore the electrical behavior of the resonant link.

Experiment A - air-gap sweep (open loop: inverter command fixed at the 15 cm design point)



Analytical air-gap sweep showing the coupling coefficient and resulting output behavior.

6. Nominal Power and Loss Design

The intended battery-side operating point is approximately 390 volt and 20.4 ampere, corresponding to 7,956 watt. For a 90 percent design-efficiency target, the required input power is approximately 8,840 watt, giving a total allowable loss

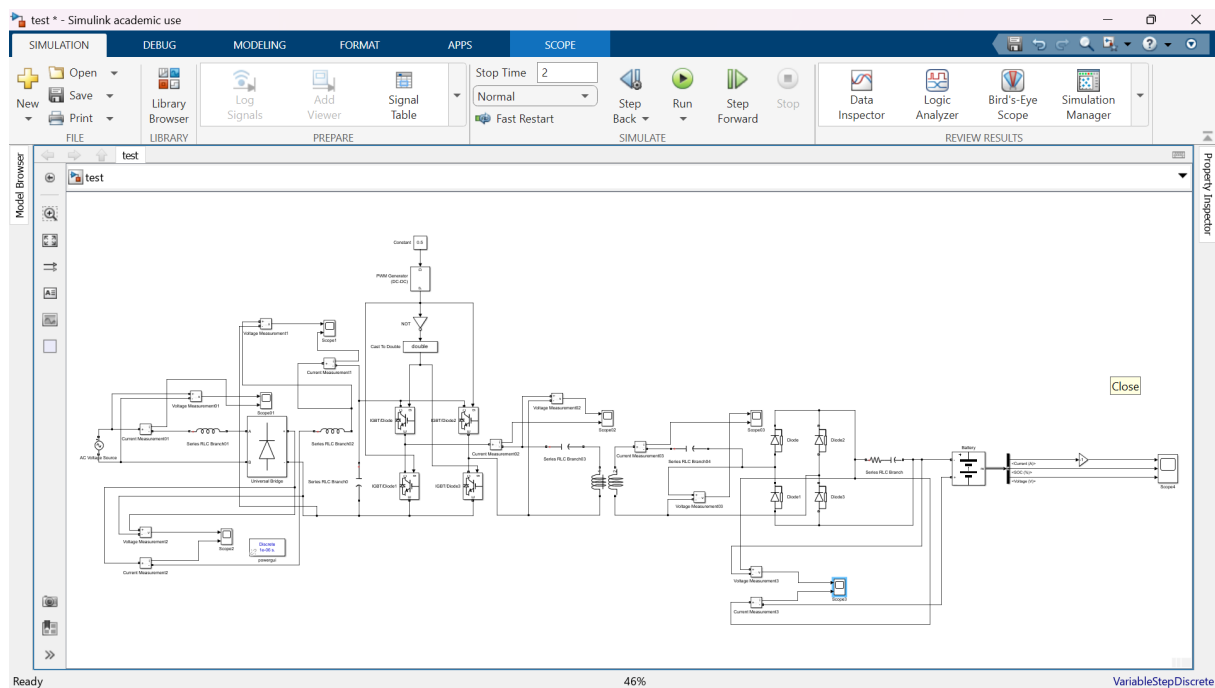
of approximately 884 watt.

Quantity	Value
Battery-side voltage	approximately 390 volt
Battery-side current	approximately 20.4 ampere
Battery-side power	7,956 watt
Target efficiency	90 percent
Target input power	approximately 8,840 watt
Total loss allowance	approximately 884 watt
Equivalent coil resistance used in the design study	approximately 0.81 ohm per coil

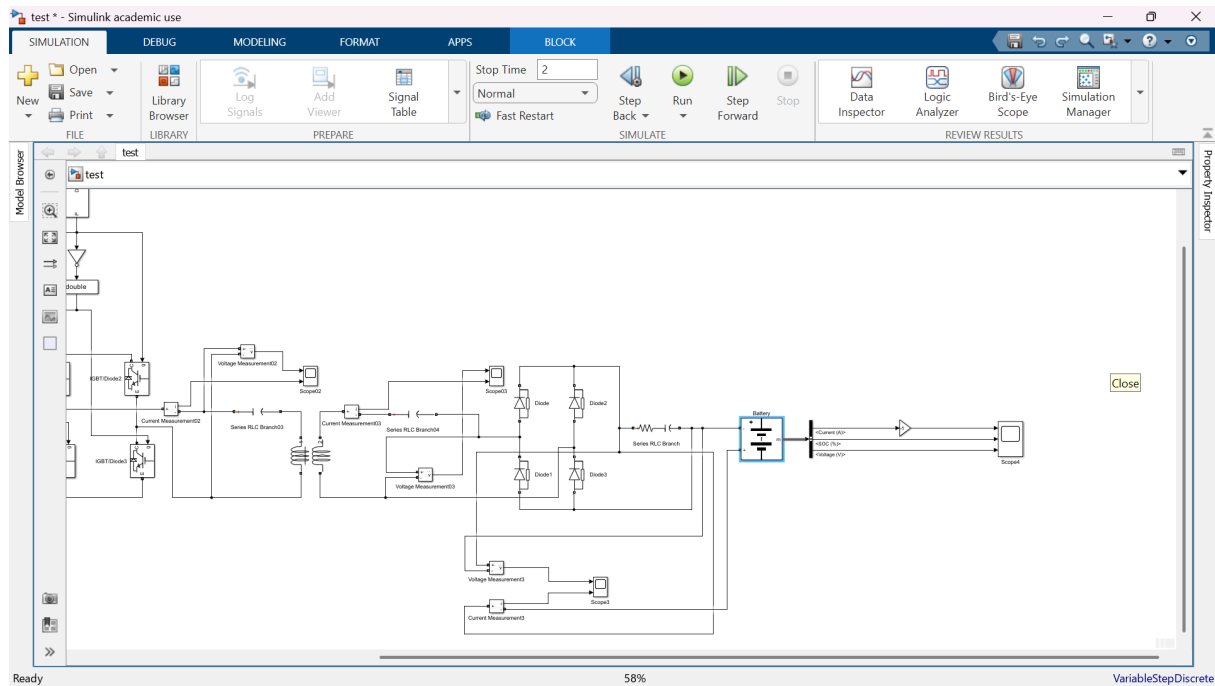
The 0.81 ohm value is an equivalent resistance derived from the loss budget. It should not be described as a measured physical coil resistance. At 30 kilohertz, the effective resistance of a real coil depends on conductor geometry, skin effect, proximity effect, winding arrangement, and magnetic materials. A future hardware-oriented version should obtain the resistance from coil geometry or direct measurement.

The existing switching-level model uses approximately 1 milliohm coil resistance. This near-ideal setting is useful for proving the wireless transfer mechanism but produces unrealistically small coil losses. Therefore, the 90 percent value is correctly presented as a design target validated by the analytical loss model, not as a measured efficiency claim from the current switching-level model.

7. Switching-Level Simulink Implementation



Full switching-level Simulink model.

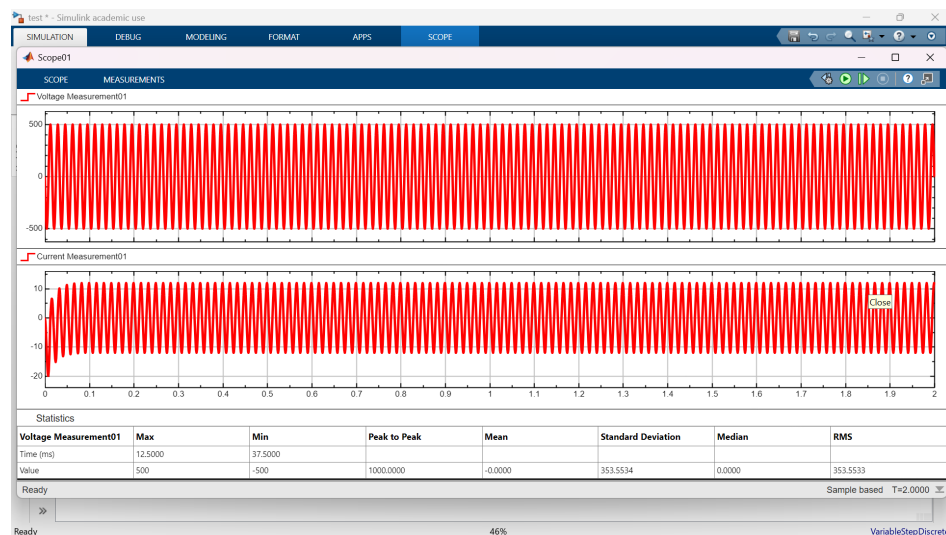


Receiver-side rectification, filtering, and battery interface.

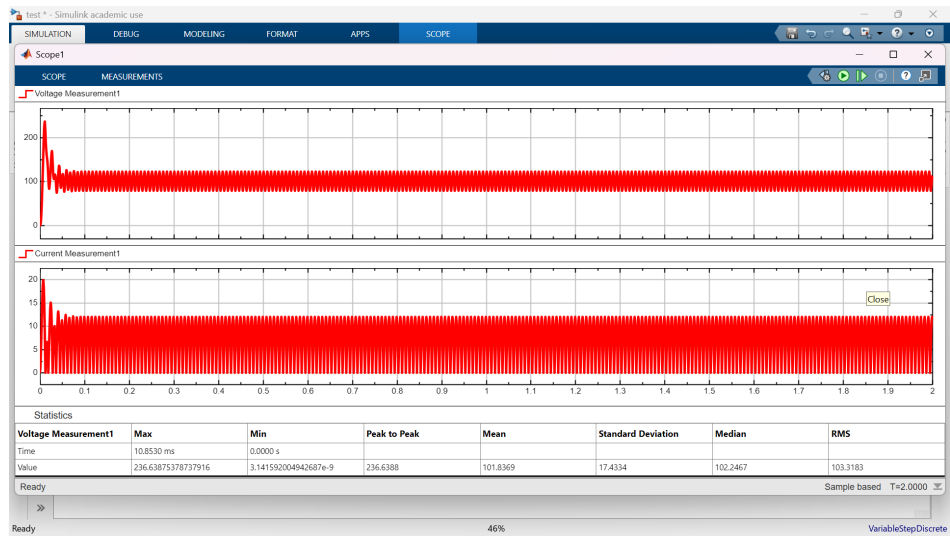
The model uses a high-frequency full-bridge inverter, resonant compensation branches, a mutual-inductance element, a diode bridge, an output filter, and a battery model. Voltage and current measurements are distributed through the chain so that the energy-transfer mechanism can be inspected rather than inferred only from the final output.

8. Switching-Level Simulation Results

The captured waveforms show the expected progression from high-frequency inverter excitation to primary resonant current, secondary-side transferred current, rectified output, filtered battery-side power, and battery charging behavior.



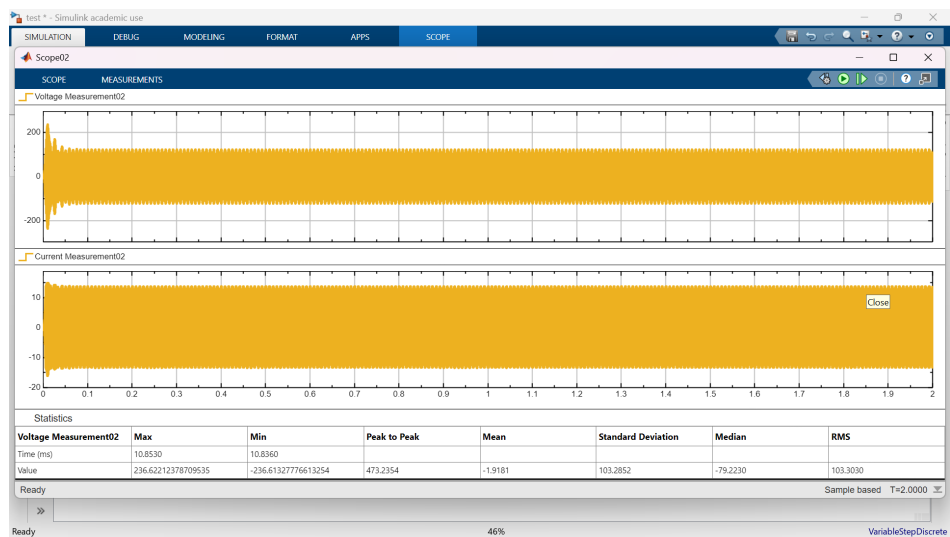
Source-side voltage and current.



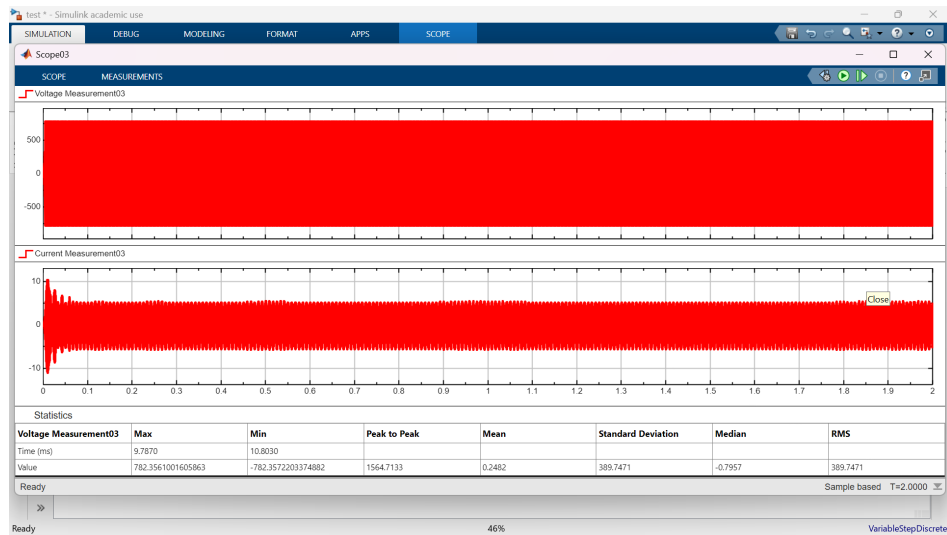
Primary resonant-side voltage and current.



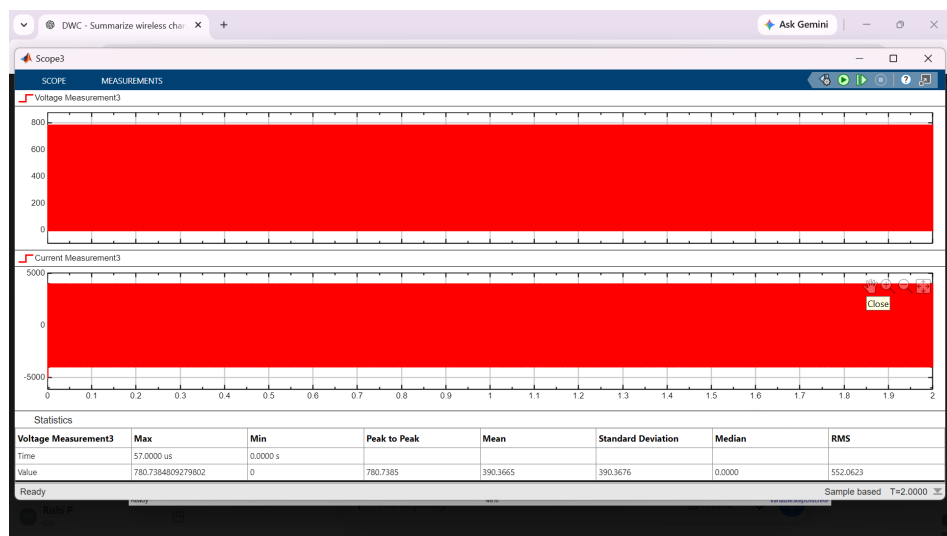
Primary compensation-network behavior.



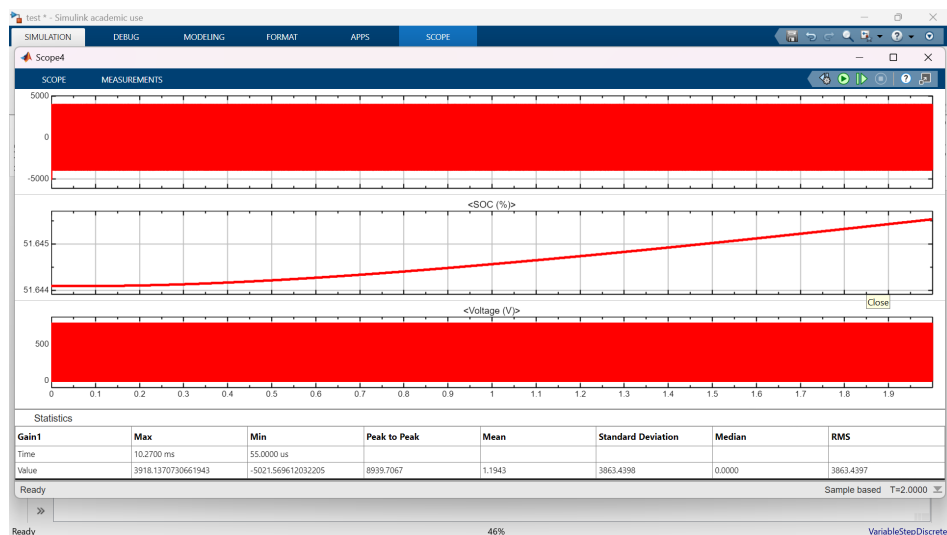
Secondary resonant-side behavior.



Rectifier-side voltage and current.



Filtered output voltage and current.



Battery current, state of charge, and terminal voltage.

9. Robustness Analysis

The separate engineering analysis treats the wireless link as a fundamental-harmonic plant so that many operating conditions can be evaluated efficiently. This is valuable because a full switching-level sweep over many air gaps, offsets, frequencies, and compensation arrangements would be computationally expensive. The analytical model is

therefore used as a design exploration tool, while the Simulink model remains the switching-level implementation.

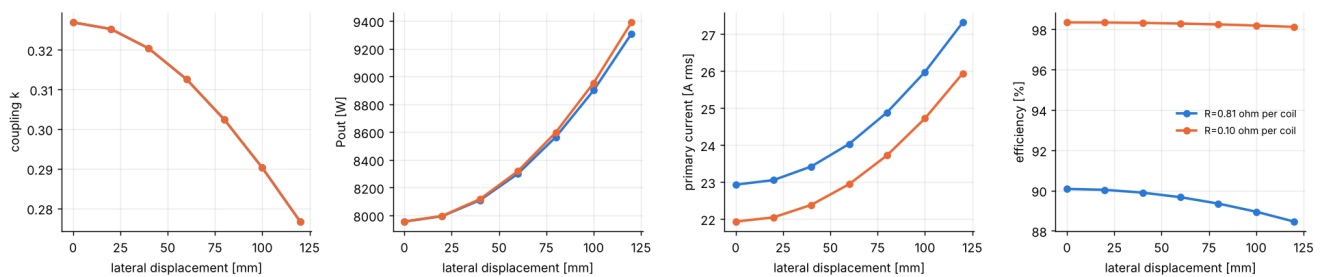
9.1 Air-gap sweep

The open-loop series-series system changes significantly as the air gap changes. Under the selected constant-voltage excitation, the calculated output power increases as coupling decreases over the studied range. This behavior is not a mistake in the model; it follows from the reflected-impedance relationship of the chosen resonant system. It also creates a control problem because a vehicle moving away from the nominal position can cause excessive power and current.

Air gap	Coupling coefficient	Output power	Efficiency
10 centimetre	0.452	5.84 kilowatt	92.6 percent
12.5 centimetre	0.382	6.86 kilowatt	91.4 percent
15 centimetre	0.327	7.96 kilowatt	90.1 percent
17.5 centimetre	0.283	9.13 kilowatt	88.7 percent
20 centimetre	0.246	10.38 kilowatt	87.2 percent
25 centimetre	0.190	13.16 kilowatt	83.8 percent

9.2 Lateral displacement

Experiment B - lateral misalignment at 15 cm gap (open loop)

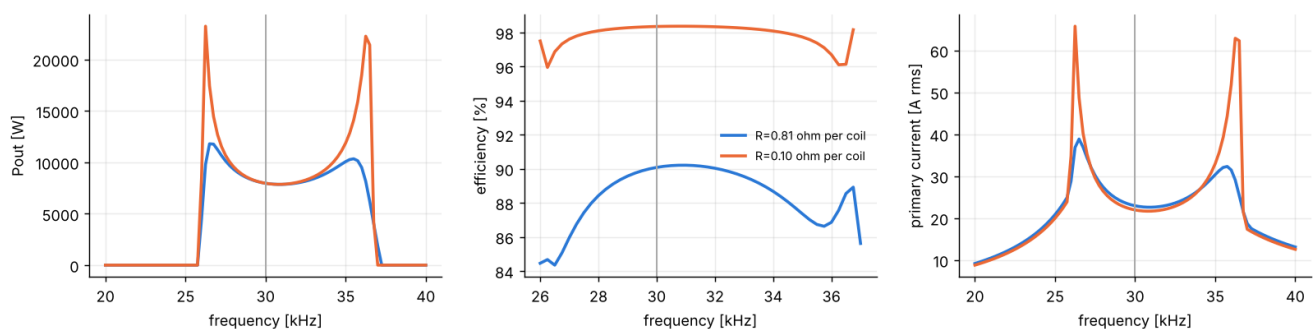


Lateral displacement sensitivity at the nominal air gap.

At a 15 centimetre air gap, the analytical model predicts a coupling coefficient reduction from approximately 0.327 to 0.302 at an 80 millimetre lateral displacement. The open-loop output changes from approximately 7.96 to 8.57 kilowatt in the equivalent-loss model. The result shows that position tolerance is a system-level issue rather than only a mechanical issue.

9.3 Frequency sensitivity

Experiment C - frequency sweep, nominal coupling (open loop, 30 kHz marked)

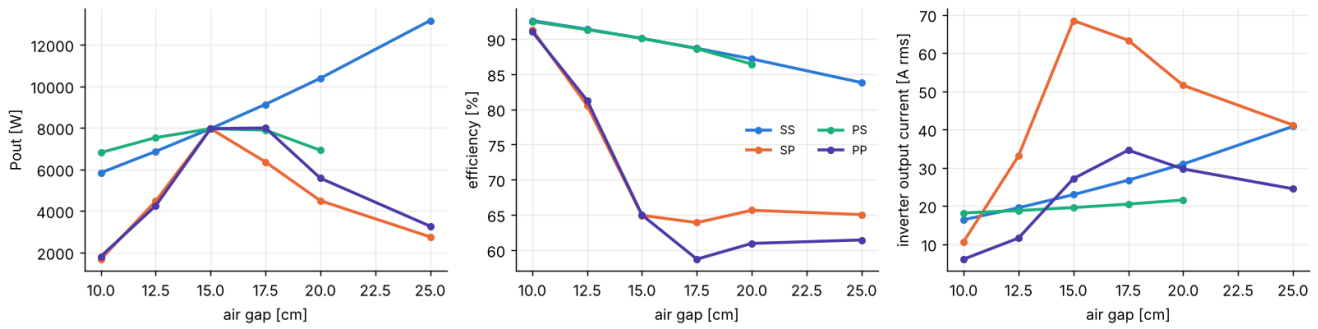


Frequency sweep around the selected resonant operating point.

The frequency study identifies a broad high-efficiency region near resonance and also reveals off-resonant regions where current can become excessive. The best calculated efficiency is close to 31 kilohertz, while 30 kilohertz remains a sound and simpler operating point because it closely matches both resonant networks.

9.4 Compensation topology comparison

Experiment D - compensation topologies, each tuned for 7.96 kW + zero input phase at 15 cm ($R = 0.81 \text{ ohm}$)



Comparison of four compensation arrangements.

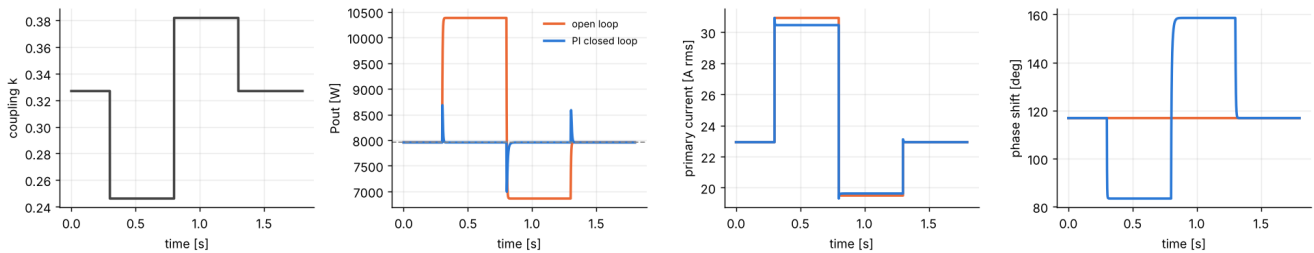
The topology comparison supports the series-series choice. At the nominal design point, series-series reaches approximately 90.1 percent efficiency with about 22.9 ampere primary current. The studied series-parallel and parallel-parallel arrangements require approximately 68.5 ampere coil current and fall to about 65 percent efficiency. The parallel-series arrangement can hold output power over a useful range but eventually loses charging capability at a sufficiently large air gap because the received voltage is no longer sufficient to forward-bias the rectifier into the battery.

10. Closed-Loop Power Regulation

The main control objective is to keep battery-side output power near 7.96 kilowatt despite changes in magnetic coupling. The proportional-integral controller compares the desired power with the measured power and changes inverter phase shift. This changes the effective power delivered by the resonant link.

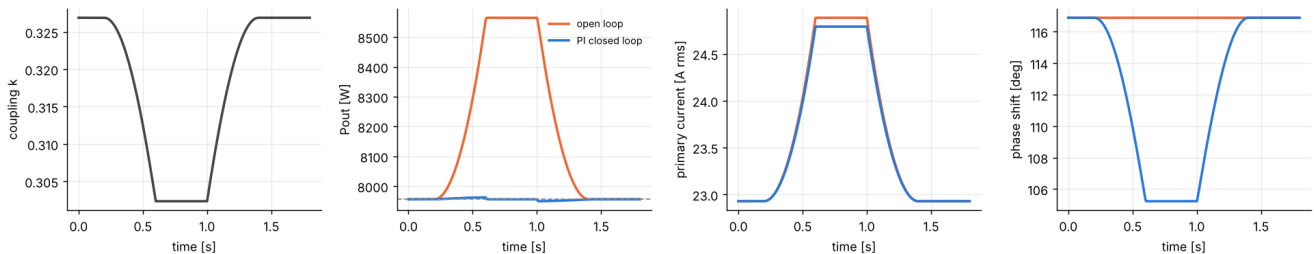
Control sequence: air gap changes → mutual inductance changes → transferred power changes → power error appears → controller changes inverter phase shift → power returns toward the reference.

E2: air-gap steps 15 → 20 → 12.5 → 15 cm (quasi-static FHA plant, $R = 0.81 \text{ ohm}$)



Air-gap step test: open-loop power changes strongly, while regulated power returns toward the 7.96 kilowatt reference.

E1: lateral misalignment 0 → 80 mm → 0 (15 cm gap) (quasi-static FHA plant, $R = 0.81 \text{ ohm}$)



Lateral displacement control test.

For the air-gap sequence of 15 centimetre to 20 centimetre to 12.5 centimetre to 15 centimetre, open-loop power reaches approximately 10.38 kilowatt at the largest gap and falls to approximately 6.86 kilowatt at 12.5 centimetre. The closed-loop analytical model holds approximately 7.96 kilowatt after each transition. The three disturbance responses return inside a ± 2 percent power band within approximately 9, 16, and 12 milliseconds.

The controller does not remove the physical cost of weak coupling. At 20 centimetre, maintaining 7.96 kilowatt requires a smaller phase shift but still produces approximately 30.5 ampere primary current and about 86.9 percent efficiency in the equivalent-loss model. At 12.5 centimetre, the required phase shift increases and the primary current falls to approximately 19.6 ampere, with about 91.3 percent efficiency. This demonstrates an important design principle: constant output power does not automatically mean constant current or constant efficiency.

11. Design Decisions and Their Rationale

Decision	Reason
30 kilohertz operating point	Provides a common target for both resonant networks and a practical switching frequency for the simulation.
Series-series compensation	Provides a direct resonant path on both coils and gives moderate current at the selected coupling.
266 microhenry transmitter coil	Sets the transmitter energy-storage characteristic and determines the required compensation capacitance.
257 microhenry receiver coil	Provides a closely matched receiver resonant characteristic for efficient magnetic energy capture.
106 and 110 nanofarad compensation	Tune the two resonant networks close to the common operating frequency.
85.46 microhenry mutual inductance	Represents the nominal magnetic coupling used by the implemented model.
0.81 ohm equivalent coil resistance in analysis	Represents the coil-loss allocation required by the 90 percent design-efficiency budget.
Phase-shift power regulation	Provides a practical control variable for limiting or increasing transferred power when coupling changes.
Air-gap and lateral-displacement sweeps	Expose sensitivity to vehicle positioning rather than evaluating only one ideal operating point.
Frequency sweep	Identifies detuning sensitivity and regions where current or power can become undesirable.

12. Engineering Interpretation

The strongest aspect of this project is not the fact that a wireless charging waveform can be produced. The stronger engineering contribution is the progression from a working physical model to a reasoned design study. The project asks what happens when the vehicle is not in the exact nominal position, why the resonant components have the selected values, why one compensation topology is preferable to another, where the loss budget goes, and how control can compensate for coupling variation.

This is the style of engineering work I aim to demonstrate: understand the physical mechanism, derive the design, implement it in a simulation environment, stress the design with realistic disturbances, measure the consequences, and clearly state what remains to be validated.

13. Limitations and Next Steps

- Replace the calibrated circular-loop magnetic model with electromagnetic simulation or measured coupling data from a realistic charging-pad geometry.
- Include ferrite, shielding, conductor skin effect, proximity effect, thermal behavior, and detailed winding geometry.
- Insert the equivalent coil resistance into the switching-level Simulink model and directly calculate averaged input and output power.
- Implement the proportional-integral controller directly around the switching-level plant rather than only the quasi-static analytical plant.
- Include sensing delay, sampling, phase-shift limits, inverter dead time, and communication delay in the control study.
- Investigate efficiency optimization jointly with power regulation so that the controller does not merely hold output power but also manages current stress and losses.
- Evaluate electromagnetic-field exposure and component thermal limits before considering physical hardware implementation.

14. Reproducibility

The repository contains the original Simulink model, parameter scripts, resonance calculations, Python analysis source, numerical result files, captured waveforms, robustness figures, and this report. The separation between switching-level simulation and analytical robustness analysis is intentional and documented so that future work can replace assumptions one by one without losing the reasoning behind the current design.

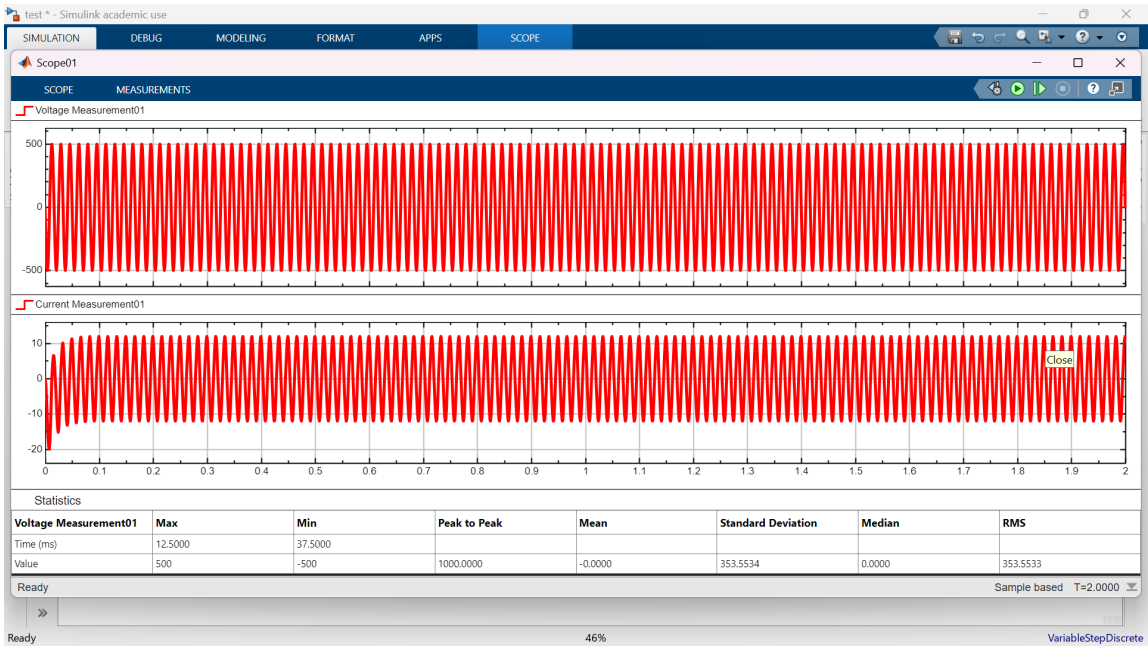
To reproduce the switching-level model, open the project in MATLAB with Simulink and Simscape Electrical available, load the parameter script, and run the model. To reproduce the analytical studies, install NumPy, SciPy, and Matplotlib and execute ``python run_experiments.py`` from the analysis directory.

15. Final Project Statement

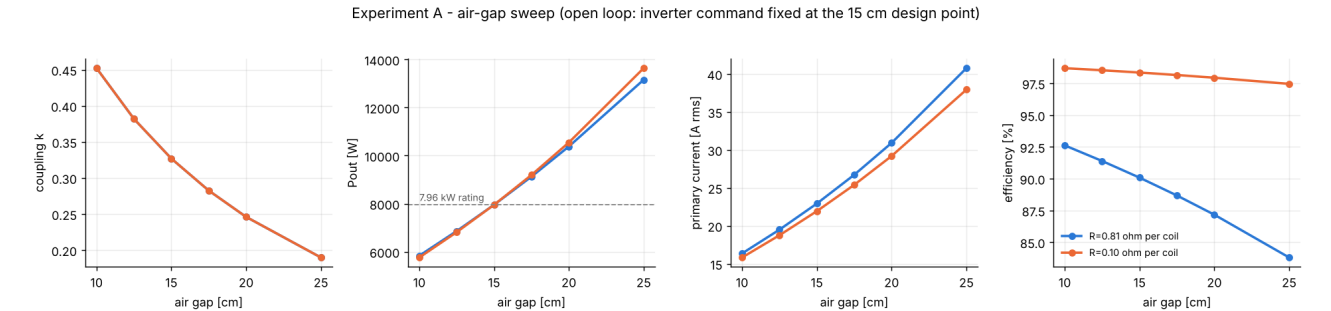
Robust Inductive Wireless Electric Vehicle Charging is a design-and-simulation project that combines resonant power electronics, magnetic coupling, wireless energy transfer, numerical design, robustness testing, and feedback control. The project demonstrates a complete wireless charging path in Simulink and then systematically investigates how the design behaves when coil separation, lateral position, and operating frequency change. The resulting work shows not only that wireless charging can be achieved in simulation, but also why the system behaves as it does and what engineering changes are required to make the design more robust and physically realistic.

Appendix A — Complete Project Figure Set

All project images are also provided separately in the ``figures`` directory so that they can be viewed directly from the repository. The following pages preserve the principal captured results in the report for convenient review.

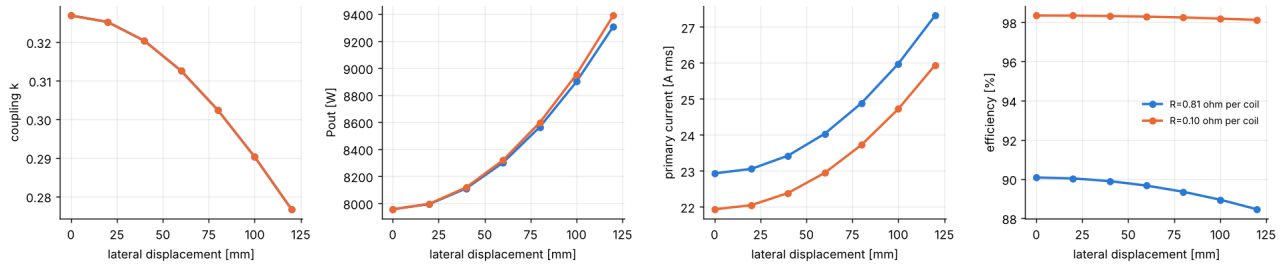


Combined output waveform capture.



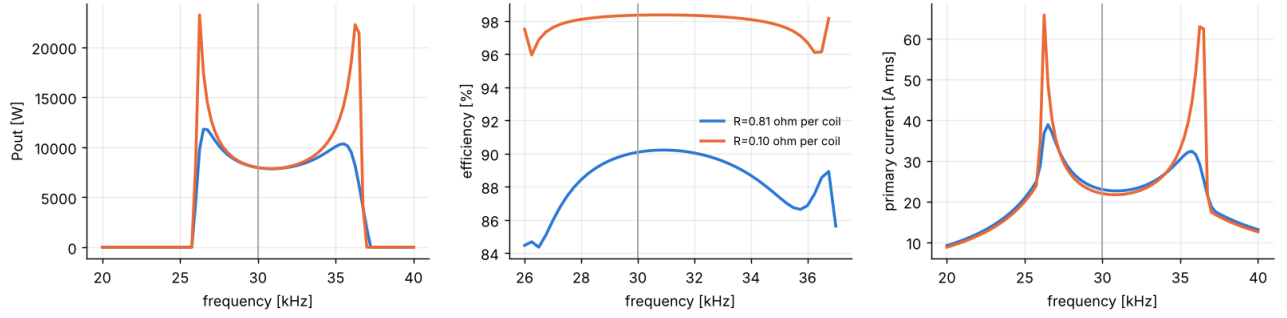
Air-gap robustness study.

Experiment B - lateral misalignment at 15 cm gap (open loop)



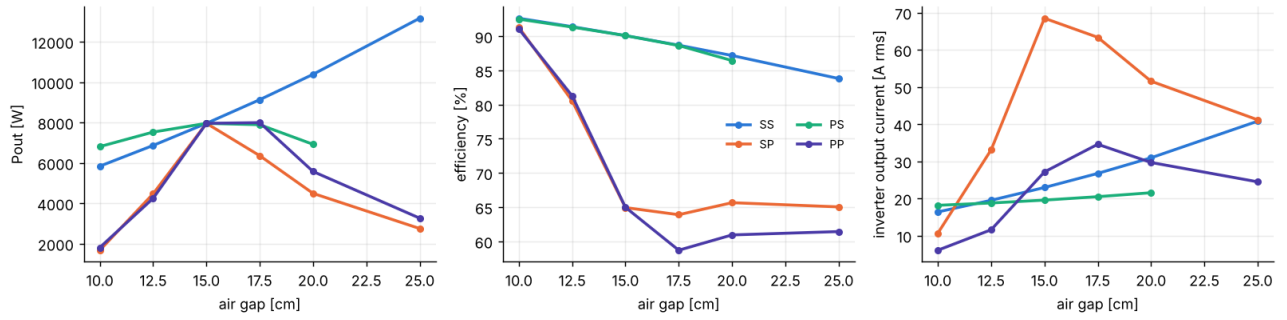
Lateral displacement study.

Experiment C - frequency sweep, nominal coupling (open loop, 30 kHz marked)



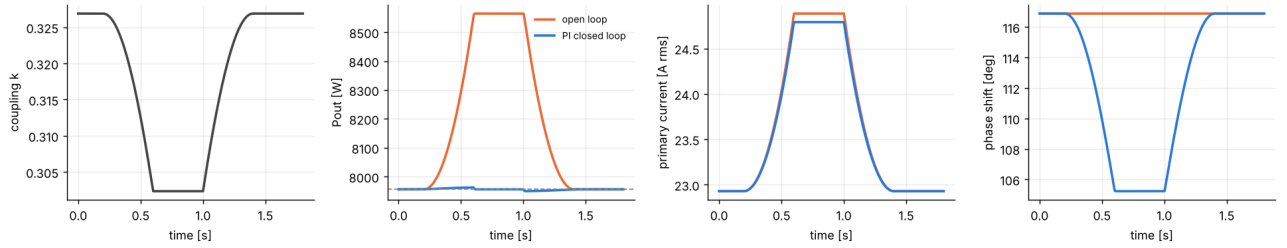
Frequency sensitivity study.

Experiment D - compensation topologies, each tuned for 7.96 kW + zero input phase at 15 cm ($R = 0.81 \text{ ohm}$)



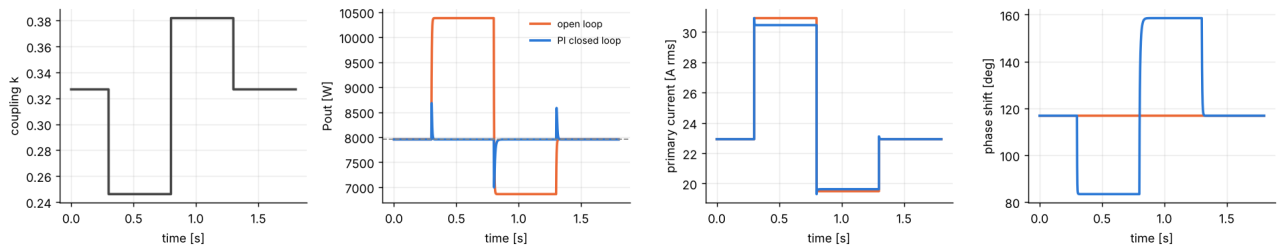
Compensation topology study.

E1: lateral misalignment 0 → 80 mm → 0 (15 cm gap) (quasi-static FHA plant, $R = 0.81 \text{ ohm}$)



Closed-loop lateral displacement study.

E2: air-gap steps 15 → 20 → 12.5 → 15 cm (quasi-static FHA plant, $R = 0.81 \text{ ohm}$)



Closed-loop air-gap step study.